

Positive Additivity and the Cauchy Step: From Vetterlein’s Euclidean Metric to Wright–Weigert’s Gleason-Type Theorem

Abstract

Vetterlein’s axiomatization of finite-dimensional Euclidean geometry ends with a uniqueness theorem for the metric. Wright and Weigert’s proof of a Gleason-type theorem begins from additive probability assignments on quantum effects and ends with the density-operator form of the Born rule. These two results belong to different mathematical worlds, but they share a precise intermediate mechanism: an additive function is first obtained on a positive domain, and positivity or boundedness rules out the pathological solutions of Cauchy’s functional equation. This note isolates that common step. In Vetterlein’s theorem it appears on each ray of Euclidean space, where $t \mapsto d(0, ta)$ is additive and nonnegative. In Wright–Weigert it appears on positive rays and cones in the real vector space of Hermitian operators, where a probability assignment is additive and bounded between 0 and 1. The analogy is not that the two theorems are the same theorem, but that both use the same regularization principle: positive additivity becomes linearity.

1 Introduction

Foundational derivations often contain a hidden analytic bottleneck. Incidence, order, or additivity assumptions first determine how a quantity behaves under composition. But additivity alone is too weak over the real numbers: without a regularity condition, Cauchy’s functional equation admits wildly discontinuous solutions. These pathologies may be constructed by choosing values on a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$; they are not continuous, not measurable, and unbounded on every interval. A proof must therefore explain why the intended physical quantity is not one of those pathological additive maps.

In Vetterlein’s final theorem, the quantity is distance. Once the synthetic geometry has been represented by a real inner-product space, the sought metric d is required to be additive along line segments and invariant under the geometric symmetries singled out by the axioms. On a fixed ray this gives an additive function

$$f_a(t) = d(0, ta), \quad t \geq 0.$$

Since d is a metric, f_a is nonnegative. This positivity is enough to force $f_a(t) = tf_a(1)$. The remaining symmetry assumptions then show that the scale $f_a(1)$ is independent of the direction of a . Thus the metric is the Euclidean metric up to a positive constant factor.

In Wright and Weigert’s article *Gleason-Type Theorems from Cauchy’s Functional Equation*, the quantity is probability. A frame function assigns numbers in $[0, 1]$ to quantum effects and is additive whenever effects can be summed inside a measurement. The effects form a positive part of the real vector space of Hermitian operators. Restricting the frame function to suitable positive rays and cones gives Cauchy-type additivity; boundedness by 0 and 1 turns this additivity into linearity; and a finite-dimensional cone-basis argument patches the local linear descriptions into

$$f(E) = \text{Tr}(\rho E)$$

for a density operator ρ .

The purpose of this note is to make the common mechanism explicit while keeping the differences clear. The Euclidean metric theorem uses a one-dimensional ray argument plus rotational symmetry. The Gleason-type theorem uses one-dimensional Cauchy regularity inside a higher-dimensional positive cone, plus a nontrivial compatibility argument across bases. In both cases, positivity supplies exactly the missing regularity.

2 The Cauchy Regularization Lemma

The elementary lemma needed for Vetterlein's metric theorem is the positive half-line version of the usual regularity results for Cauchy's equation.

Definition 1. *In this note, a function is called positively additive on a positive domain if it satisfies Cauchy's equation wherever the sum is defined and is bounded below or bounded above on that domain. In the applications below, this boundedness is supplied by the physical interpretation of the quantity: distance is bounded below by zero, while probability is bounded between zero and one.*

Lemma 1 (positive Cauchy lemma). *Let $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ satisfy*

$$f(s+t) = f(s) + f(t)$$

for all $s, t \geq 0$. If $f(t) \geq 0$ for all $t \geq 0$, then

$$f(t) = tf(1)$$

for all $t \geq 0$.

Proof. First $f(0) = 0$. Positivity implies monotonicity: if $0 \leq s \leq t$, then

$$f(t) = f(s) + f(t-s) \geq f(s).$$

For $n \in \mathbb{N}$, additivity gives $f(n) = nf(1)$, and hence

$$f\left(\frac{m}{n}\right) = \frac{m}{n}f(1)$$

for nonnegative rationals m/n . If $t \geq 0$, choose rationals q, r with $q \leq t \leq r$. Monotonicity gives

$$qf(1) = f(q) \leq f(t) \leq f(r) = rf(1).$$

Letting $q, r \rightarrow t$ yields $f(t) = tf(1)$. □

Lemma 2 (finite-interval Cauchy lemma). *Let $a \geq 1$, and let $f : [0, a] \rightarrow \mathbb{R}$ satisfy*

$$f(x+y) = f(x) + f(y)$$

whenever $x, y, x+y \in [0, a]$. If f is bounded above or bounded below on $[0, a]$, then

$$f(x) = xf(1), \quad x \in [0, a].$$

Proof. This is a standard result in the theory of functional equations; see, for example, Aczel–Dhombres or the boundedness condition in Wright–Weigert's Theorem 1. The bounded-below case follows from the bounded-above case by applying the bounded-above result to $-f$. □

Remark 1. *The point is that boundedness on an interval supplies the regularity that Cauchy additivity alone does not provide.*

3 Vetterlein's Metric Theorem and the Ray Cauchy Equation

The last theorem in Vetterlein's paper asserts the uniqueness, up to a positive scale factor, of the metric compatible with the Euclidean geometry derived in the preceding sections. After the representation theorem, one may describe the relevant part of the argument in the following linear model.

Proposition 1 (metric uniqueness in the linear model). *Let V be a real Euclidean vector space of dimension at least 3, with Euclidean norm $\|\cdot\|$. Let d be a metric on V satisfying:*

(M1) *if b lies between a and c , then*

$$d(a, c) = d(a, b) + d(b, c);$$

(M2a) *d is translation invariant:*

$$d(x + v, y + v) = d(x, y);$$

(M2b) *d is invariant under every orthogonal map with a nonzero fixed vector:*

$$d(Tx, Ty) = d(x, y).$$

Then there is a constant $\lambda > 0$ such that

$$d(x, y) = \lambda \|x - y\|$$

for all $x, y \in V$.

Proof. By translation invariance, $d(x, y) = D(y - x)$, where $D(v) = d(0, v)$. Fix $a \in V$, $a \neq 0$, and define

$$f_a(t) = D(ta) = d(0, ta), \quad t \geq 0.$$

For $s, t \geq 0$, the point sa lies on the segment from 0 to $(s + t)a$. Thus (M1) gives

$$f_a(s + t) = d(0, (s + t)a) = d(0, sa) + d(sa, (s + t)a).$$

Using translation invariance on the second term,

$$d(sa, (s + t)a) = d(0, ta) = f_a(t),$$

and hence

$$f_a(s + t) = f_a(s) + f_a(t).$$

Since d is a metric, $f_a(t) \geq 0$. The positive Cauchy lemma therefore gives

$$D(ta) = f_a(t) = tD(a), \quad t \geq 0. \tag{1}$$

It remains to show that $D(a)$ depends only on the Euclidean length of a . Let u, w be unit vectors. Since $\dim V \geq 3$, there is an orthogonal operator T sending u to w and fixing some nonzero vector.¹ By (M2b),

$$D(w) = d(0, w) = d(T0, Tu) = d(0, u) = D(u).$$

Thus all Euclidean unit vectors have the same d -length. Write this common value as λ . Because d is a metric, $\lambda > 0$. For any $v \neq 0$, write $v = \|v\|u$ with u a unit vector. Equation (1) gives

$$D(v) = D(\|v\|u) = \|v\|D(u) = \lambda\|v\|.$$

Finally,

$$d(x, y) = D(y - x) = \lambda\|y - x\|.$$

□

¹This is one place where the dimension assumption matters. In dimension 2, for example, a rotation sending u to $-u$ fixes no nonzero vector.

This proof isolates the Cauchy step in Vetterlein's final theorem. Segment additivity gives the functional equation on a ray. Translation invariance identifies all translated copies of the same ray. Positivity of the metric excludes the non-linear additive solutions. Orthogonal invariance then removes directional dependence.

One can summarize the flow as follows:

$$\begin{array}{c}
\text{betweenness additivity} + \text{translation invariance} \\
\Downarrow \\
t \mapsto d(0, ta) \text{ is additive on } \mathbb{R}_{\geq 0} \\
\Downarrow \quad \text{positivity of } d \\
d(0, ta) = t d(0, a) \\
\Downarrow \quad \text{orthogonal invariance} \\
d(x, y) = \lambda \|x - y\|.
\end{array}$$

4 Wright–Weigert: Additivity on Effect Cones

We now transport the same analytic strategy to the quantum setting. The ray becomes a positive operator ray; the distance becomes a probability assignment. Wright and Weigert work with effects, which represent unsharp yes–no questions or individual outcomes of generalized quantum measurements. Let $\mathcal{E}_d$ be the set of effects on $\mathbb{C}^d$, that is, Hermitian operators E satisfying

$$0 \leq E \leq I.$$

The real vector space in the background is

$$\text{Herm}_d = \{A = A^* : A \in M_d(\mathbb{C})\},$$

equipped with the trace inner product

$$\langle A, B \rangle = \text{Tr}(AB).$$

A frame function is a map

$$f : \mathcal{E}_d \rightarrow [0, 1]$$

with $f(I) = 1$, satisfying additivity whenever the sum remains an effect:

$$f(E_1 + E_2) = f(E_1) + f(E_2), \quad E_1, E_2, E_1 + E_2 \in \mathcal{E}_d.$$

Busch's Gleason-type theorem says that such an f must be of the form

$$f(E) = \text{Tr}(\rho E)$$

for a density operator ρ . Busch and others established this effect-valued Gleason theorem by different methods; Wright and Weigert's contribution, for our purposes, is to make explicit a proof route through Cauchy's functional equation.

The Cauchy analogy becomes visible by restricting f to positive rays. If B is a fixed effect, then for suitable $x \in [0, a]$ the operator xB is again an effect. Define

$$F_B(x) = f(xB).$$

Whenever $x, y, x + y \in [0, a]$, additivity of f gives

$$F_B(x + y) = F_B(x) + F_B(y).$$

Moreover $0 \leq F_B(x) \leq 1$. Wright–Weigert's finite-interval Cauchy theorem therefore implies

$$F_B(x) = xF_B(1)$$

on that interval. This is the same regularization step as in Vetterlein: an additive function on a positive domain becomes linear because it is bounded.

The quantum theorem, however, requires more than ray-linearity. The domain $\mathcal{E}_d$ is not a vector space but a convex positive body inside the vector space $\mathbf{Herm}_d$. Wright and Weigert use augmented bases and positive cones to extend the one-dimensional Cauchy conclusion to finite-dimensional linearity. For a fixed augmented basis $B = \{B_1, \dots, B_{d^2}\}$, ray-linearity gives linear behavior on positive combinations

$$E = \sum_j x_j B_j, \quad x_j \geq 0,$$

whenever these combinations remain effects. A priori, a different augmented basis could yield a different linear functional. The nontrivial part of the Wright–Weigert proof is to use overlaps of positive cones, together with minimal informationally complete measurements, to force these local linear descriptions to agree on enough effects to span the full real vector space of Hermitian operators. Very schematically:

$$\begin{array}{c} \text{additivity of probabilities on coexistent effects} \\ \Downarrow \\ x \mapsto f(xB_j) \text{ satisfies Cauchy's equation on intervals} \\ \Downarrow \quad f : \mathcal{E}_d \rightarrow [0, 1] \\ f(xB_j) = xf(B_j) \\ \Downarrow \quad \text{positive-cone and basis patching} \\ f(E) = \langle c, E \rangle = \text{Tr}(\rho E). \end{array}$$

The final identification of the representing vector with a density operator uses positivity and normalization: $f(E) \geq 0$ on effects forces the representing operator to be positive semidefinite, and $f(I) = 1$ gives trace one.

5 The Analogy and Its Limits

The common principle is best stated as follows:

Additivity determines rational behavior. Positivity or boundedness supplies the regularity needed to extend rational behavior to real linearity.

In Vetterlein's theorem the relevant additive maps are scalar functions on rays:

$$t \mapsto d(0, ta).$$

The role of positivity is played by the nonnegativity of a metric.

In Wright–Weigert's theorem the relevant additive maps are restrictions of a probability assignment to positive rays and positive cones in an operator space:

$$x \mapsto f(xB), \quad E \mapsto f(E).$$

The role of positivity is played by the codomain condition $f(E) \in [0, 1]$ and ultimately by the positivity of quantum probabilities.

This also explains why the full richness of Gleason's theorem is not encountered in Vetterlein's metric theorem. The point is not simply that Vetterlein works with real Euclidean spaces rather than complex Hilbert spaces. Real Hilbert spaces also have Gleason-type phenomena. The decisive difference is the domain on which additivity is imposed. In the metric theorem, additivity is tested on one-dimensional rays and then transported between rays by Euclidean symmetry:

Cauchy on rays + orthogonal invariance.

In Gleason-type theorems, additivity is imposed on a highly interlocking family of projections or effects, equivalently on many overlapping measurement contexts. The problem is not only to make each one-dimensional restriction linear, but to prove that all those local linear descriptions are compatible with a single global linear functional. We can summarize the structural difference in slogan form:

Vetterlein: *Cauchy on rays plus Euclidean symmetry,*

Gleason: *Cauchy/additivity on a web of effects plus context compatibility.*

The dimension bound $n \geq 3$ therefore plays different roles in the two settings. In Vetterlein it supplies enough affine/projective geometry and orthogonal symmetry for the reconstruction of Euclidean space. In Gleason’s original projection theorem it is tied to the overlap structure of measurement contexts; dimension two is exceptional for projections, though effect-valued versions recover Gleason-type conclusions there.

There are also important differences. Vetterlein’s metric theorem is a uniqueness theorem for a distance once the Euclidean affine and orthogonal structure has already been reconstructed. Its Cauchy step is one-dimensional and elementary. Wright–Weigert’s theorem is a representation theorem for states on an operator effect space. Its Cauchy step is still one-dimensional locally, but the main work lies in extending those local linearities across a family of cones so that a single global linear functional results.

Thus the two arguments should not be identified wholesale. Rather, they share a structural lemma. Positivity prevents additive quantities from being Hamel pathologies. After that, the geometry-specific or quantum-specific symmetries finish the representation.

6 Conclusion

Vetterlein’s last theorem and Wright–Weigert’s Gleason-type theorem both pass through Cauchy’s functional equation. In the Euclidean case, the metric axioms give an additive nonnegative function on each ray, and hence linear dependence on the scalar parameter. In the quantum case, additivity of probabilities gives Cauchy equations on positive operator directions, and boundedness of probabilities forces linearity there. The final forms

$$d(x, y) = \lambda \|x - y\| \quad \text{and} \quad f(E) = \text{Tr}(\rho E)$$

are then obtained by adding the relevant invariance or compatibility principles.

The conceptual bridge is therefore not merely the word “additivity”. It is the stronger pattern of *positive additivity*: an additive law on a positive domain, together with nonnegativity or boundedness, forces the linear structure that the physical interpretation requires. In this sense the relevant linear structure is not simply postulated at the final stage: it is recovered from additivity once the physical positivity of distance or probability has ruled out the pathological Cauchy solutions.

References

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